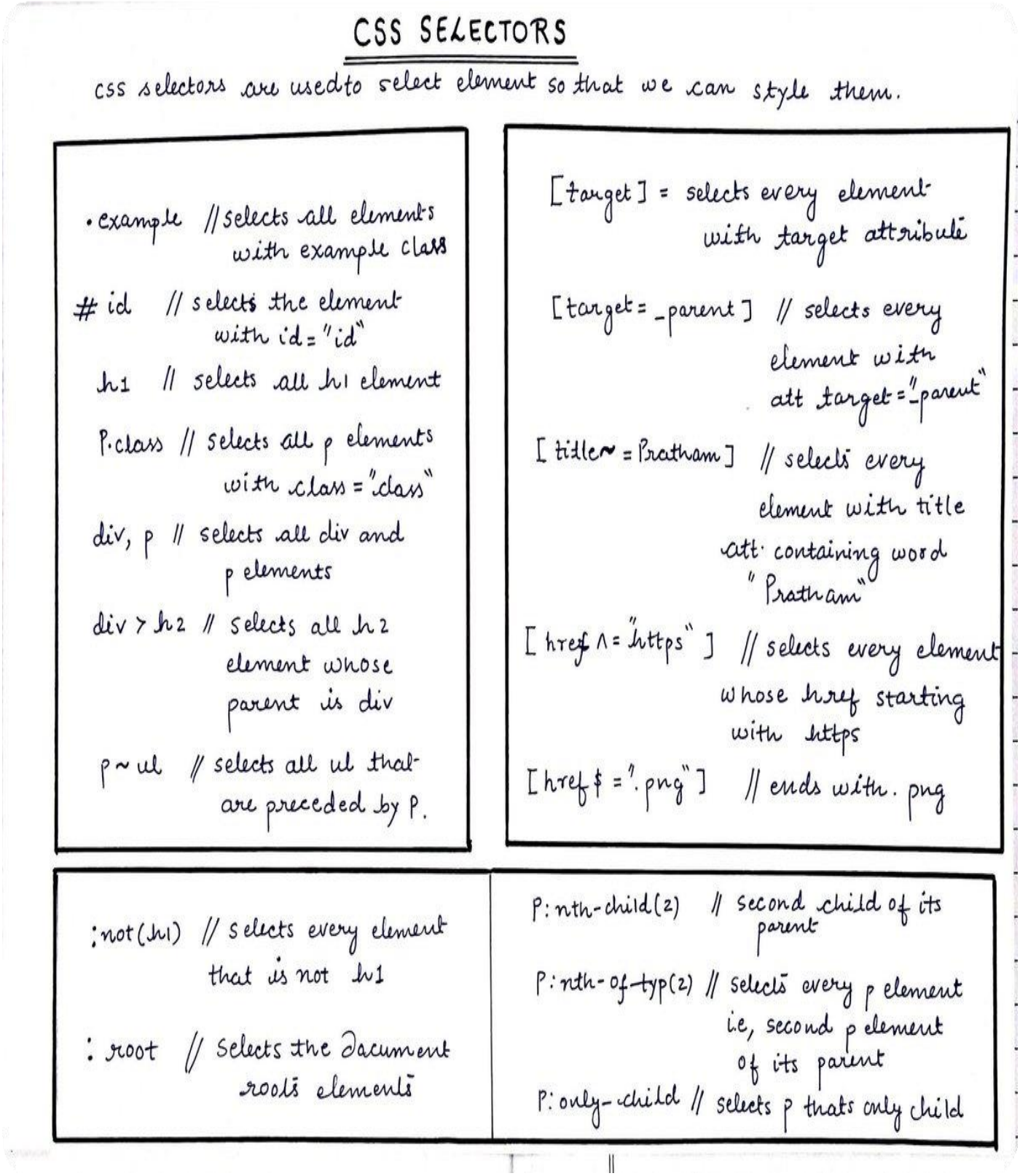
1. CSS selectors



2. Some important CSS Pseudo classes

CSS Pseudo Classes.

Selector	Example	Description.
:active	a:active	Selects the active link
:checked	input:checked	selects every checked input element.
:enabled	input:enabled	Selects every enabled input element
:empty	p:empty	Selects every p elements that has no children.
:first-child	p:first-child	selects every p elements that is the first child of its parent.
:first-of-type	p:first-of-type	Selects every p element that is the first p element of its parent.
:focus	input:focus	selects the input element that has focus.
:hover	a:hover	Selects a on mouse over.
:in-range	input:in-range	selects input elements with a value within a specified range.
:not(selector)	:not(p)	Selects all element ~~ex~~ except p.
:nth-child	p:nth-child(2)	Selects every p elements that is second child of its parent.
:only-of-type	p:only-of-type	Selects every p elements that is the only p element of its parent.
:optional	input:optional	Selects input element with no required attribute.

3. Box model

CSS BOX-MODEL

- Everything in CSS is a box or rather everything in HTML is a box-model which is surrounded by 4 different box virtually.

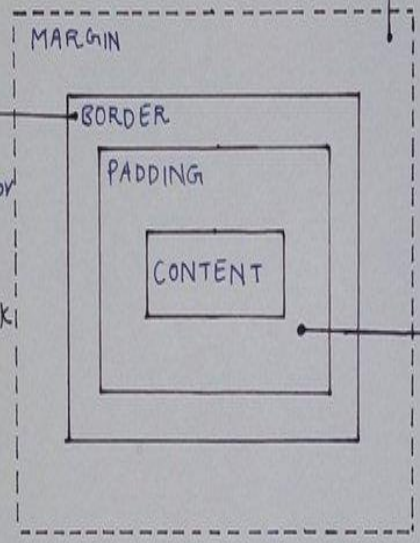
1. Content (original content inside element)
2. Padding (create space b/w content and element's border)
3. Border (create border around element)
4. Margin (space between element)

→ Margin defined the space between element.
for ex. margin: 10px;
It will create 10px empty space around element in all direction.

margin: 25px 5px 6px 10px;
↓ ↓ ↓ ↓
top right bottom left

margin: 25px 10px 25px
// Top, right and left, bottom

margin: 25px, 10px; and
// Top, bottom, right, left

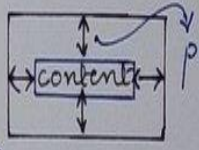


you can create border around element by specifying the width, color and style.

for ex: border: 1px solid black;

dotted:
dashed:
double:
Solid:

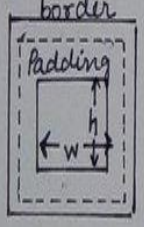
→ Padding allows you to create space between content and element's boundary for ex.



padding: 10px;

// similarly as margin, you can pass four, three, two or one value in padding as well.

you can write padding: 10%;
%- specify a padding or margin in % of the containing element.



If you add width as 100px, padding as 10px and border as 2px then the entire width becomes 112px (100+10+2).

Box-sizing: border-box;

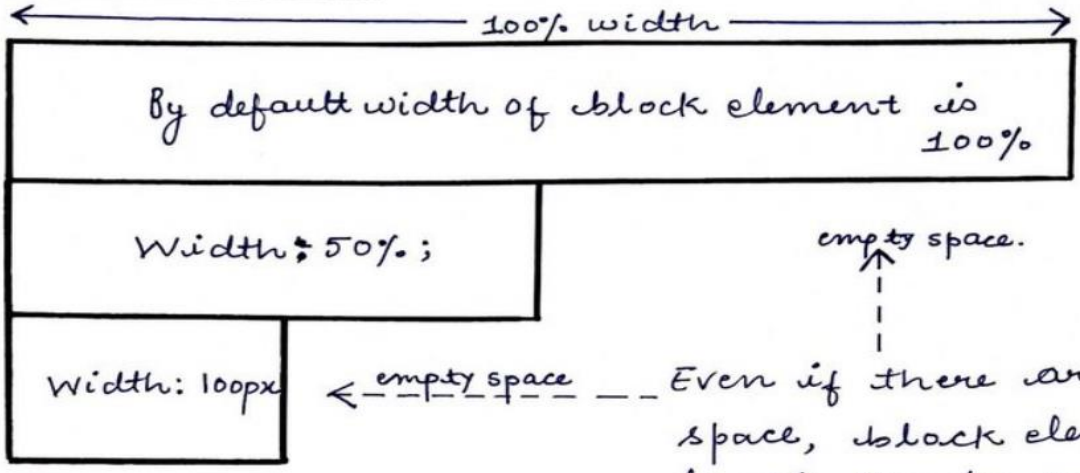
The box-sizing property defines how the width and height of an element are calculated:
if we apply box-sizing: border-box then the padding and border will be adjusted in the width and height of an element.

PRATHAM ☺

4. Display property

Display : Block, inline and inline-block.

Block Elements.

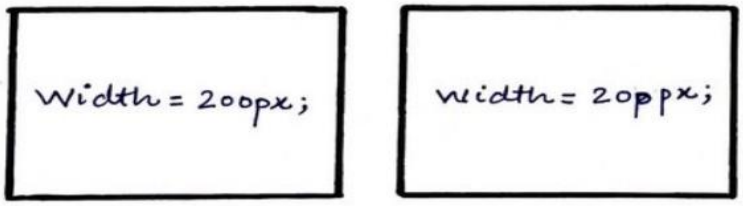


Inline

You can't control width and height of inline element. for ex. span.

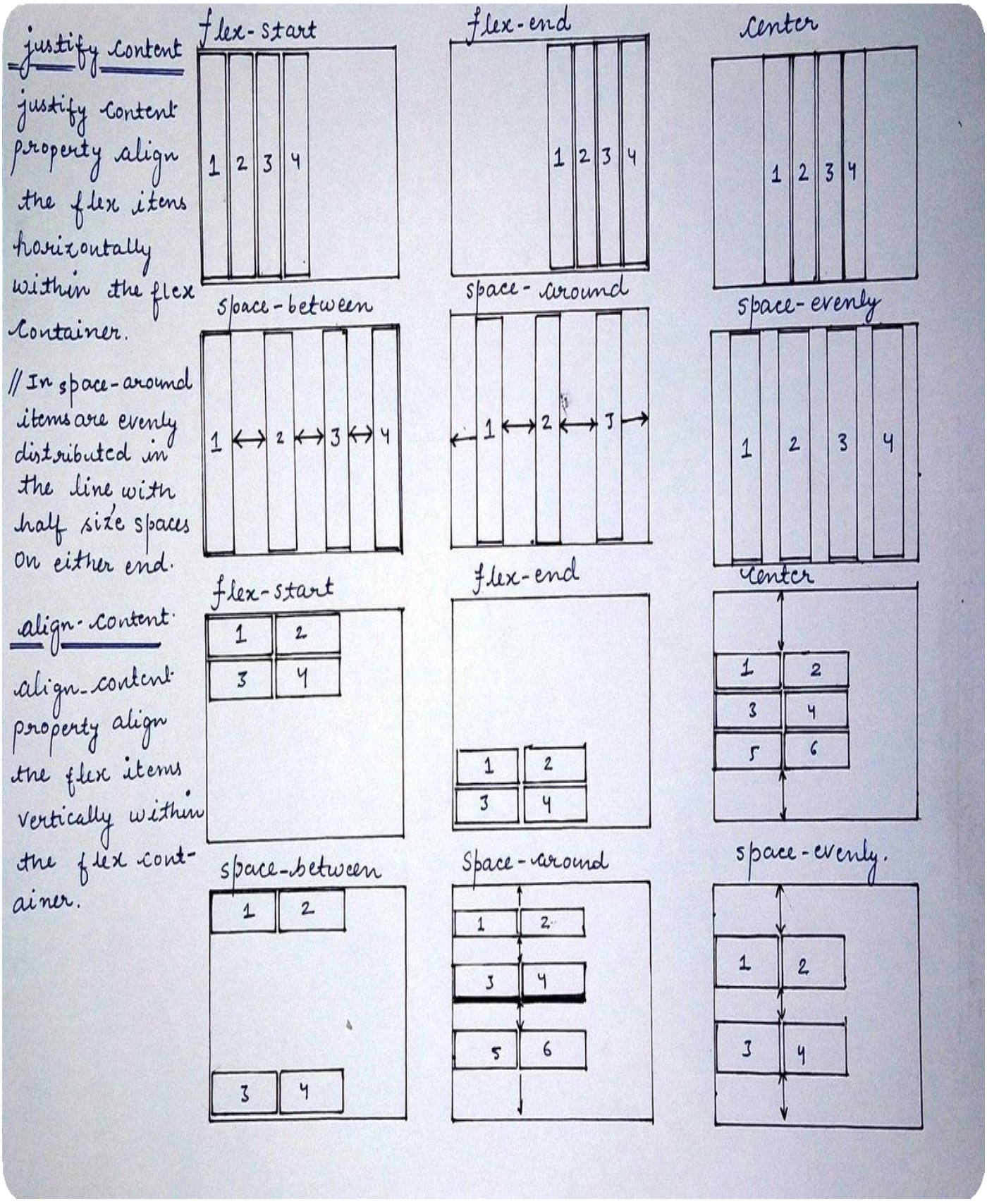
Inline-block.

(As the term suggest, they are combination of block and inline)

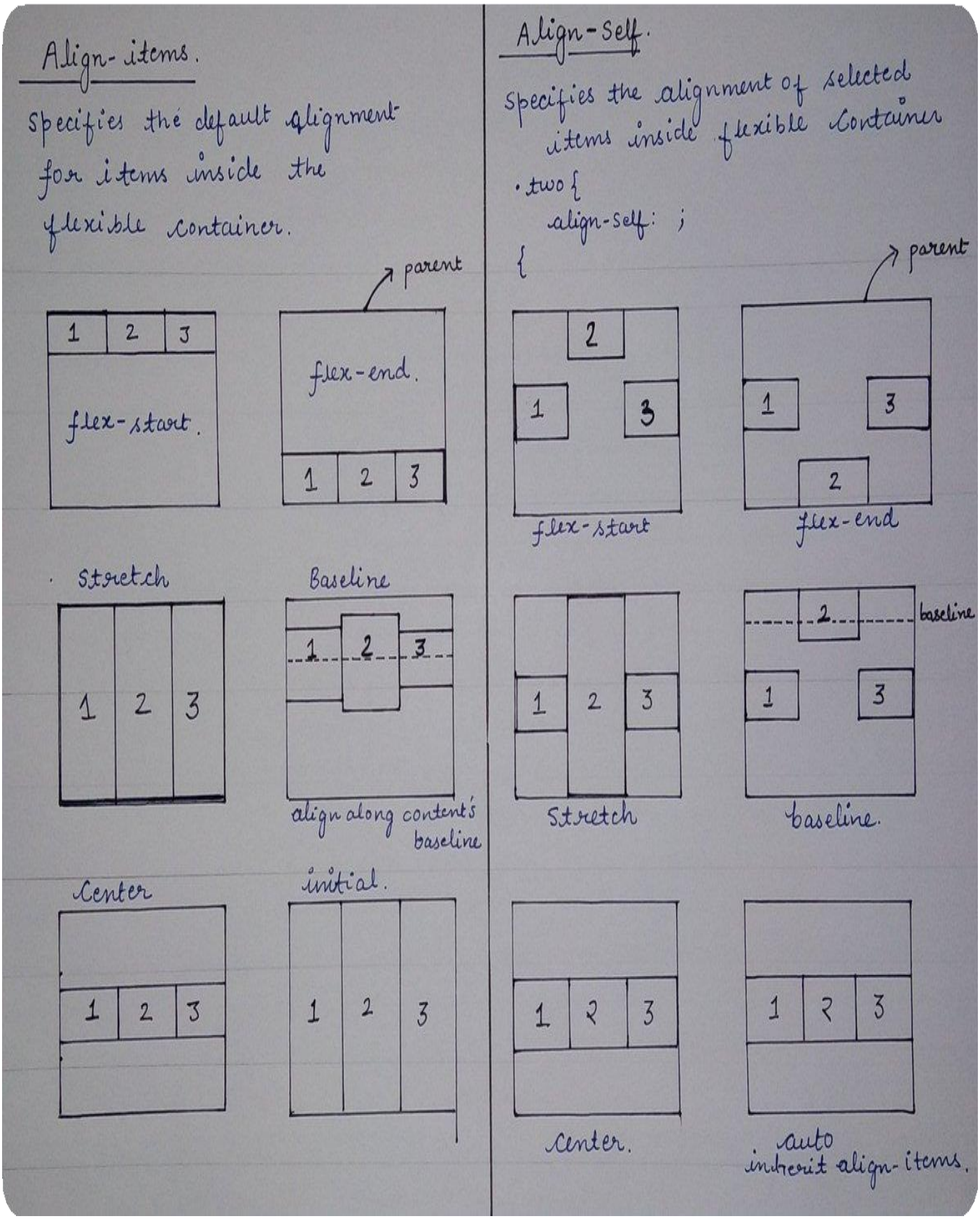


The element is formatted as inline element but you can apply height and width as well.

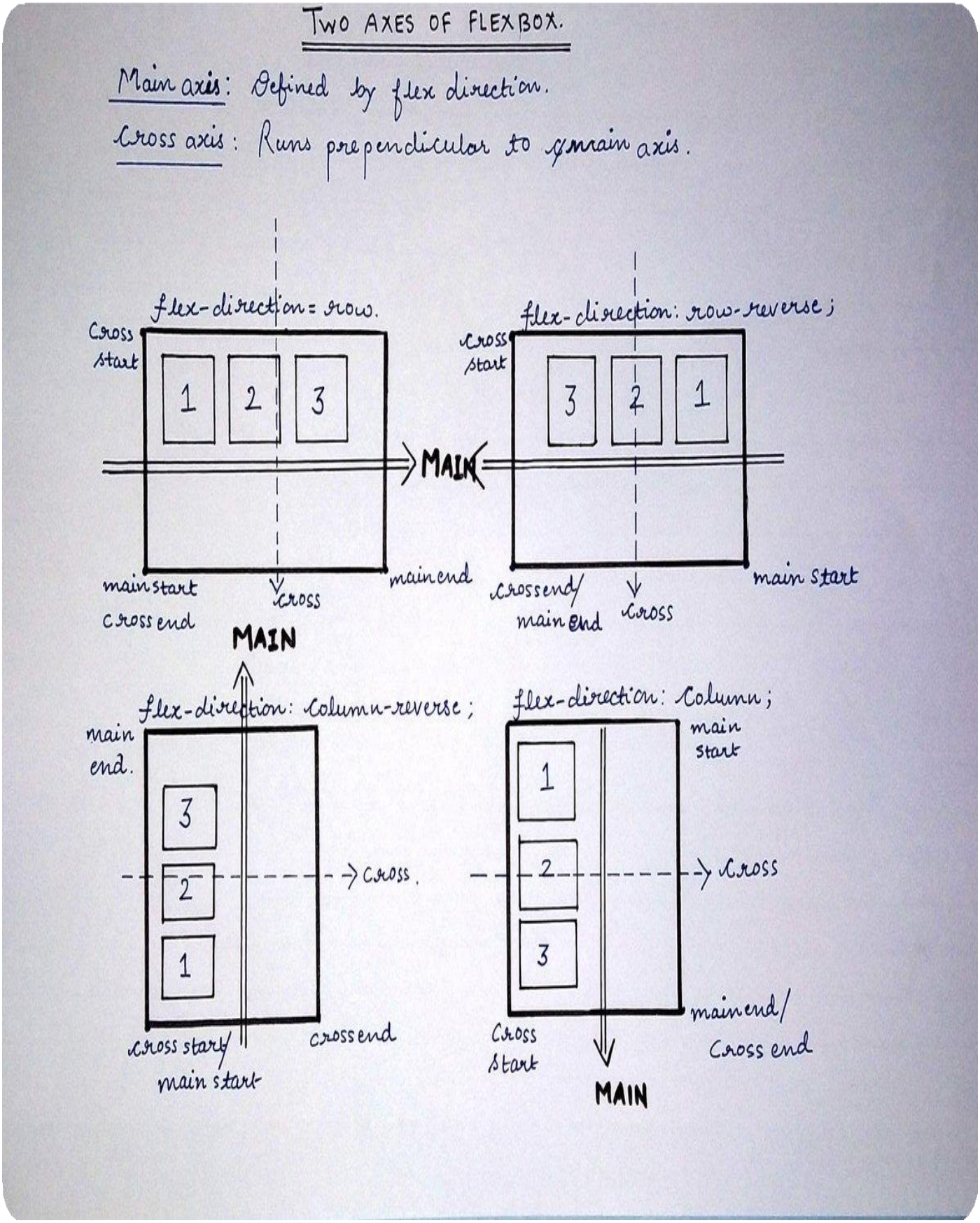
5. Justify-content and align-content explained



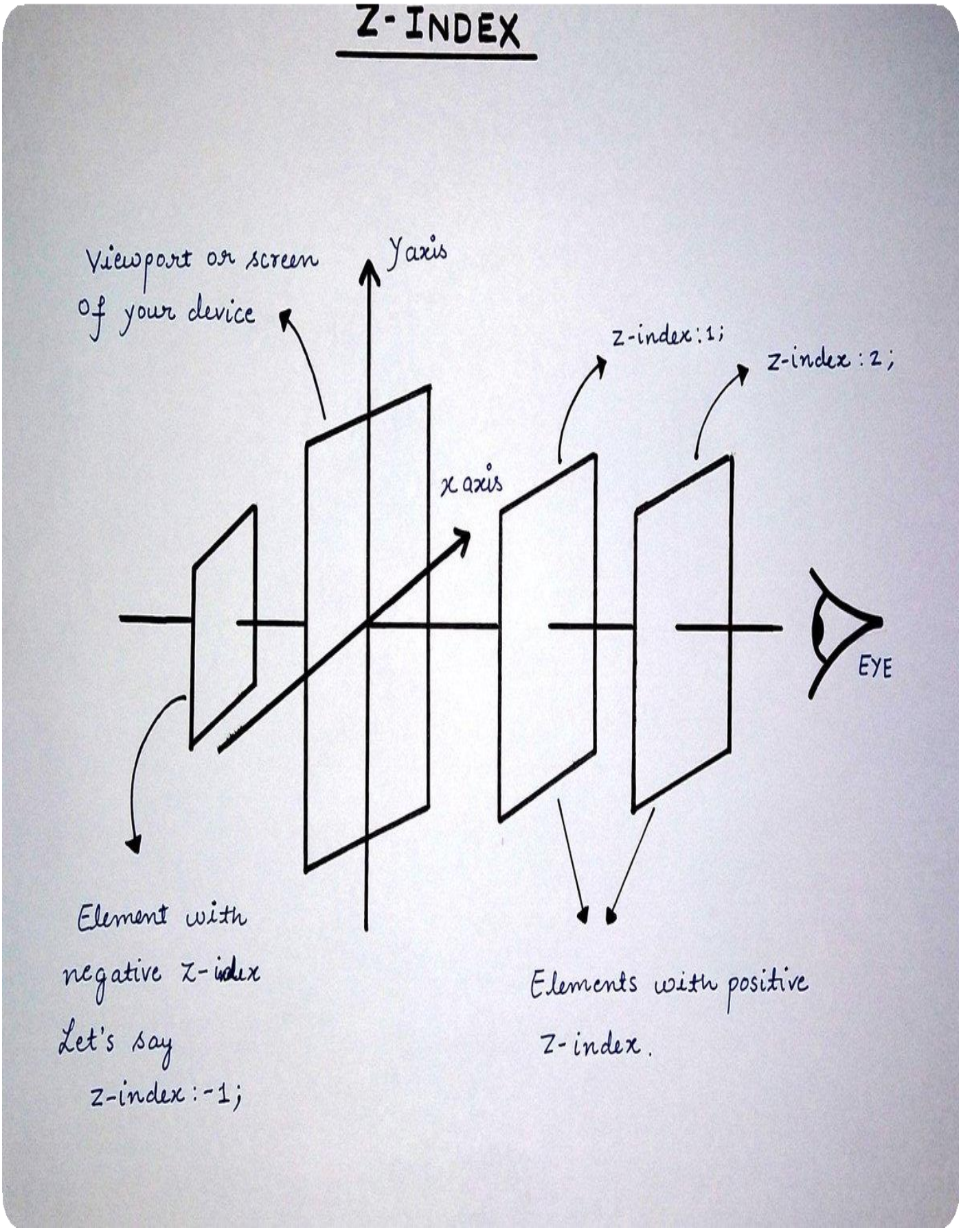
6. Align-items and align-self explained



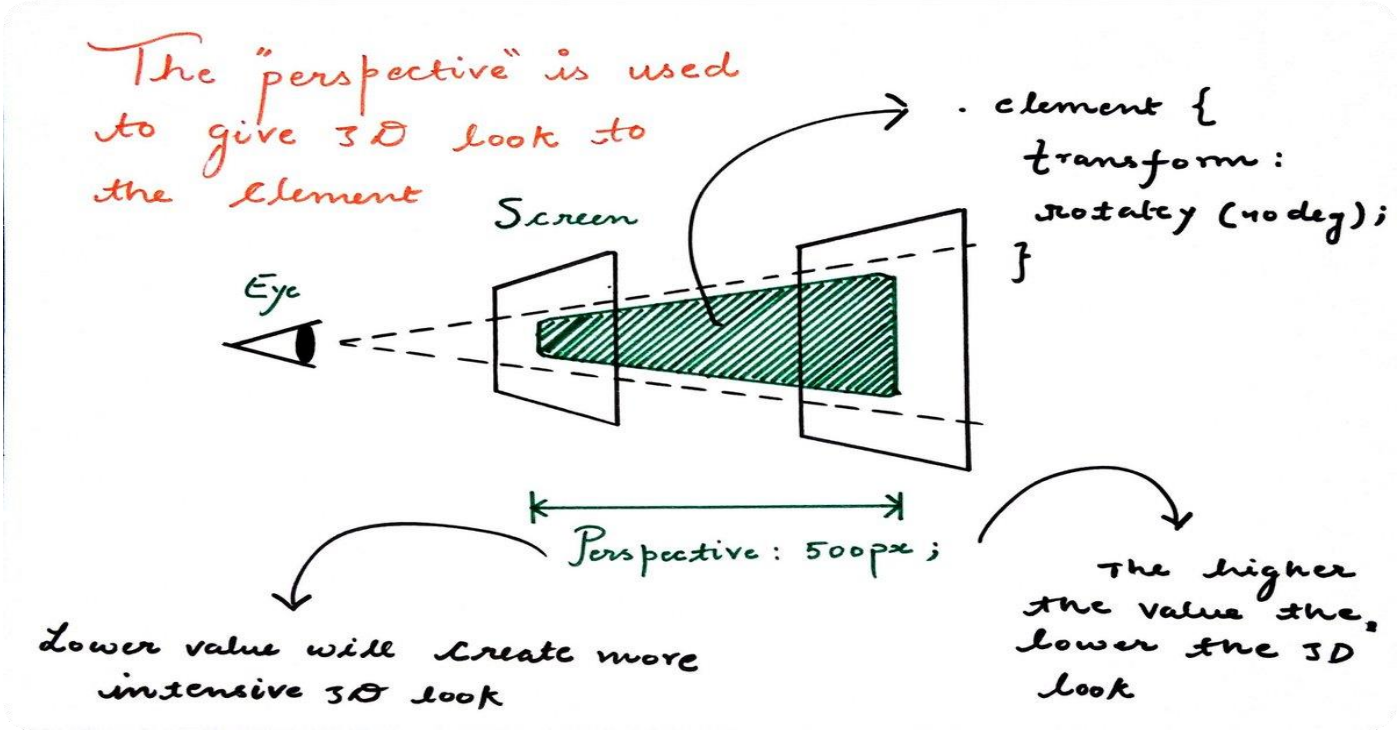
7. Axes of a layout container

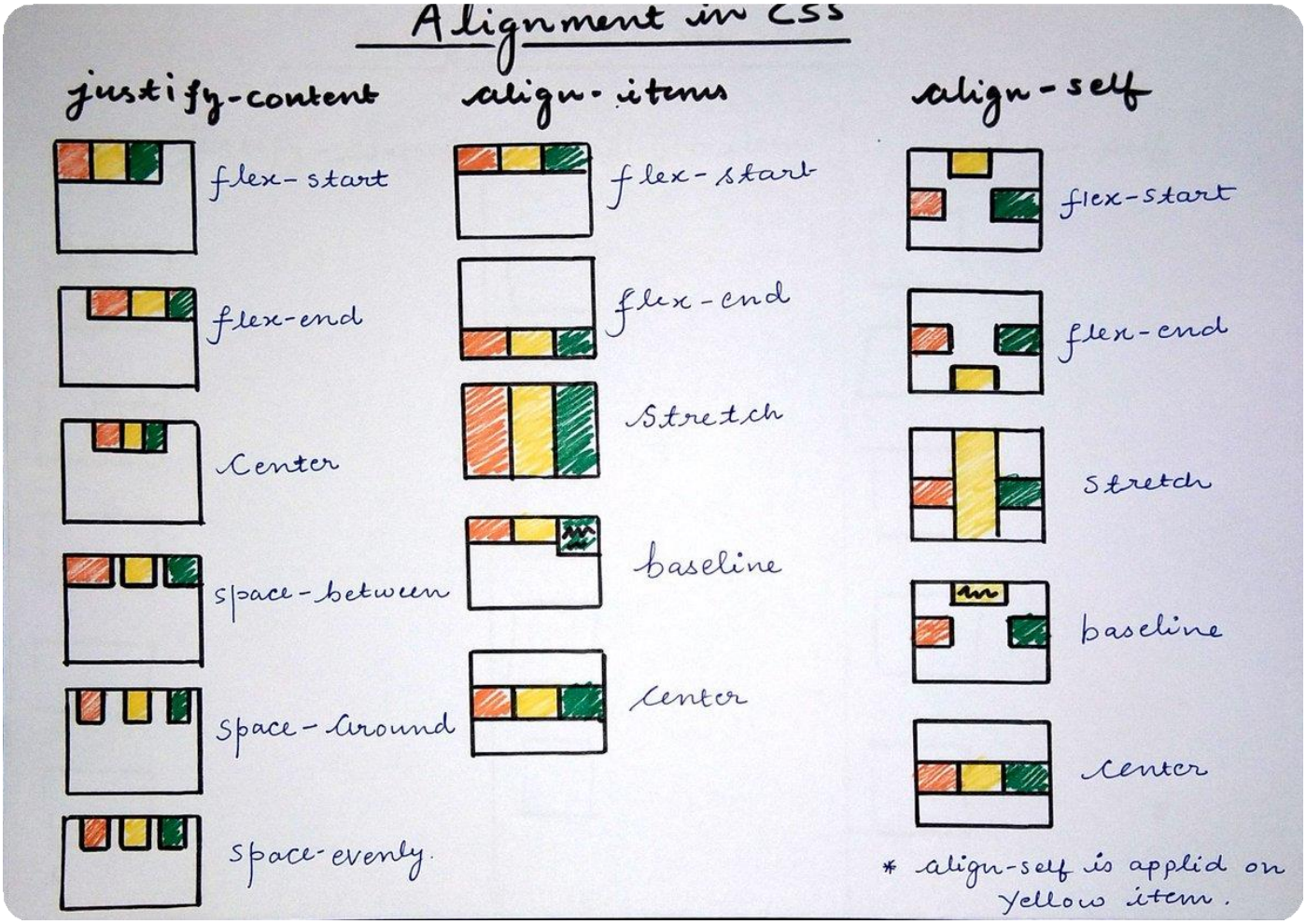


8. z-index - stacking order of elements



9. Give your element a 3D look wit1h CSS perspectiveproperty





Gird Layout

```
.one {
  grid-column: 1/4;
}
```

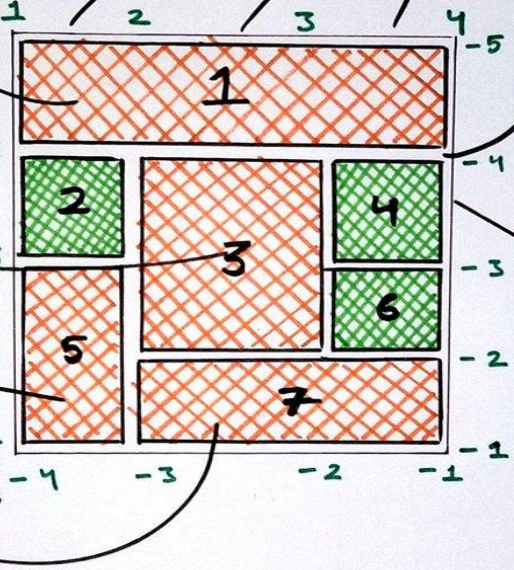
```
.three {
  grid-column: 2/3;
  grid-row: 2/4;
}
```

```
.five {
  grid-row: 3/5;
}
```

```
.seven {
  grid-column: 2/4;
}
```

grid-template-column: 1fr 2fr 1fr

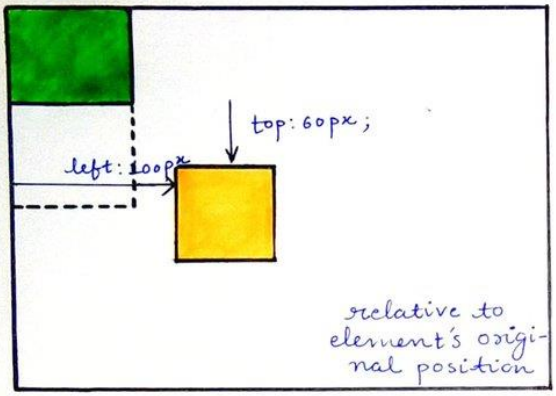
grid-gap: 10px;




```
.parent {
  justify-content: center;
  align-content: center
}
```

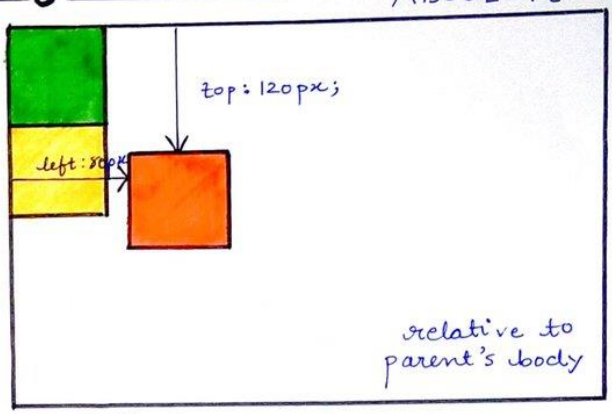
Positioning in Css

RELATIVE



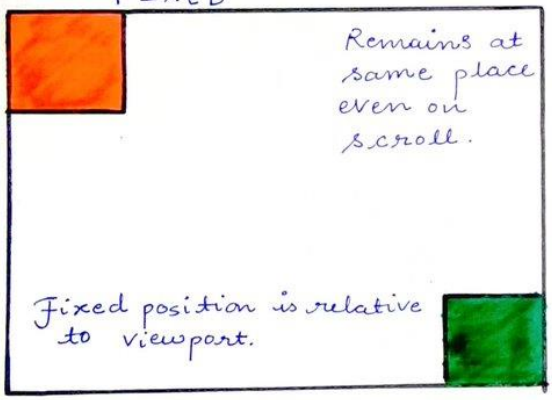
relative to element's original position

ABSOLUTE



relative to parent's body

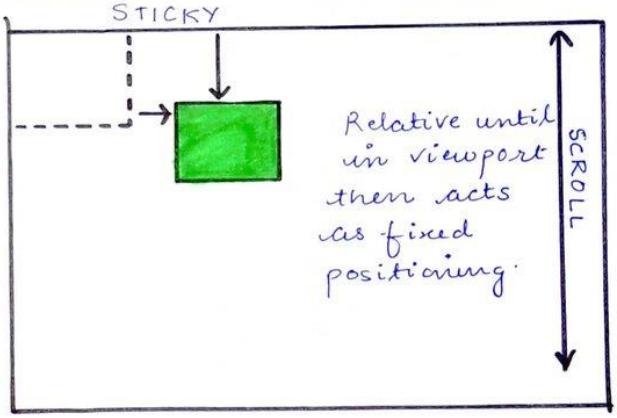
FIXED



Remains at same place even on scroll.

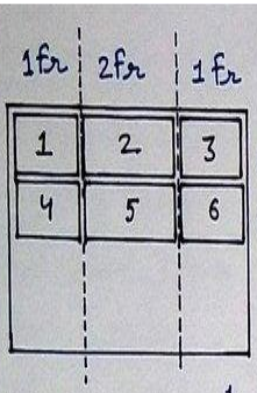
Fixed position is relative to viewport.

STICKY

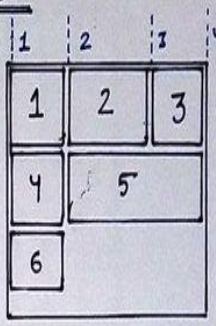


Relative until in viewport then acts as fixed positioning.

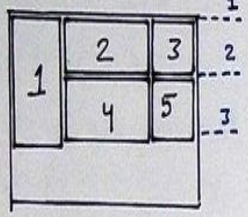
Grid cheat sheet.



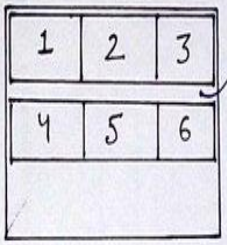

```
grid-gap: 10px;
grid-template-column: 1fr 2fr 1fr;
grid-gap: 10px;
display: grid;
```



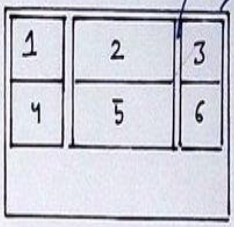

```
.five {
  grid-column: 2/4;
}
// fifth element start from 2nd column and ends at 4th column.
```




```
.one {
  grid-row: 1/3;
}. // row gap ⟶
```



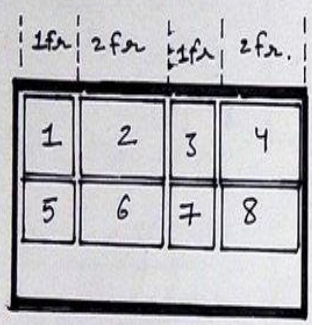
grid-row-gap: 10px



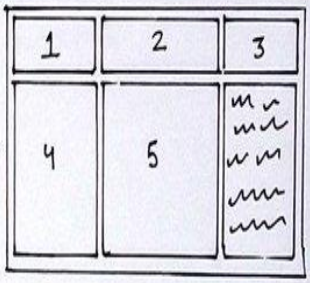
grid-column-gap: 10px;

// column gaps.

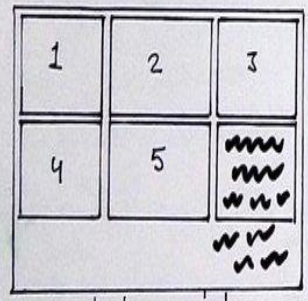
// we can use grid-gap. It will add equal space b/w rows and column.



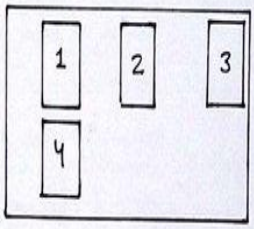

```
grid-template-column:
  repeat(2, 1fr 2fr);
// It will repeat column 2 times as 1 fraction an two fraction
```



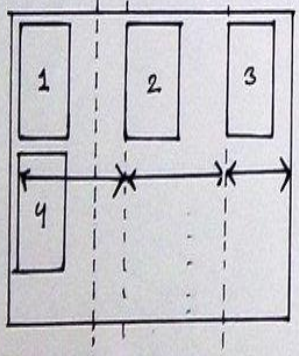

```
grid-auto-column:
  minmax(100px, auto);
// min height 100px and max will be auto.
```



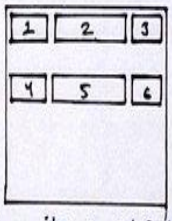

```
.parent {
  grid-auto-row: 100px;
}
// every item has 100px height. Due to which content overflows. In order to prevent it.
```



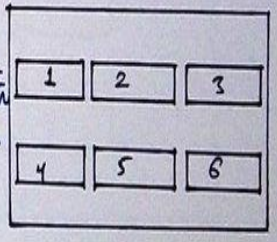

```
.parent { justify-items: end; }
// it will take center value as well. which will align the items center horizontally.
```




```
.parent {
  justify-items: start;
}
// it will align horizontally.
```

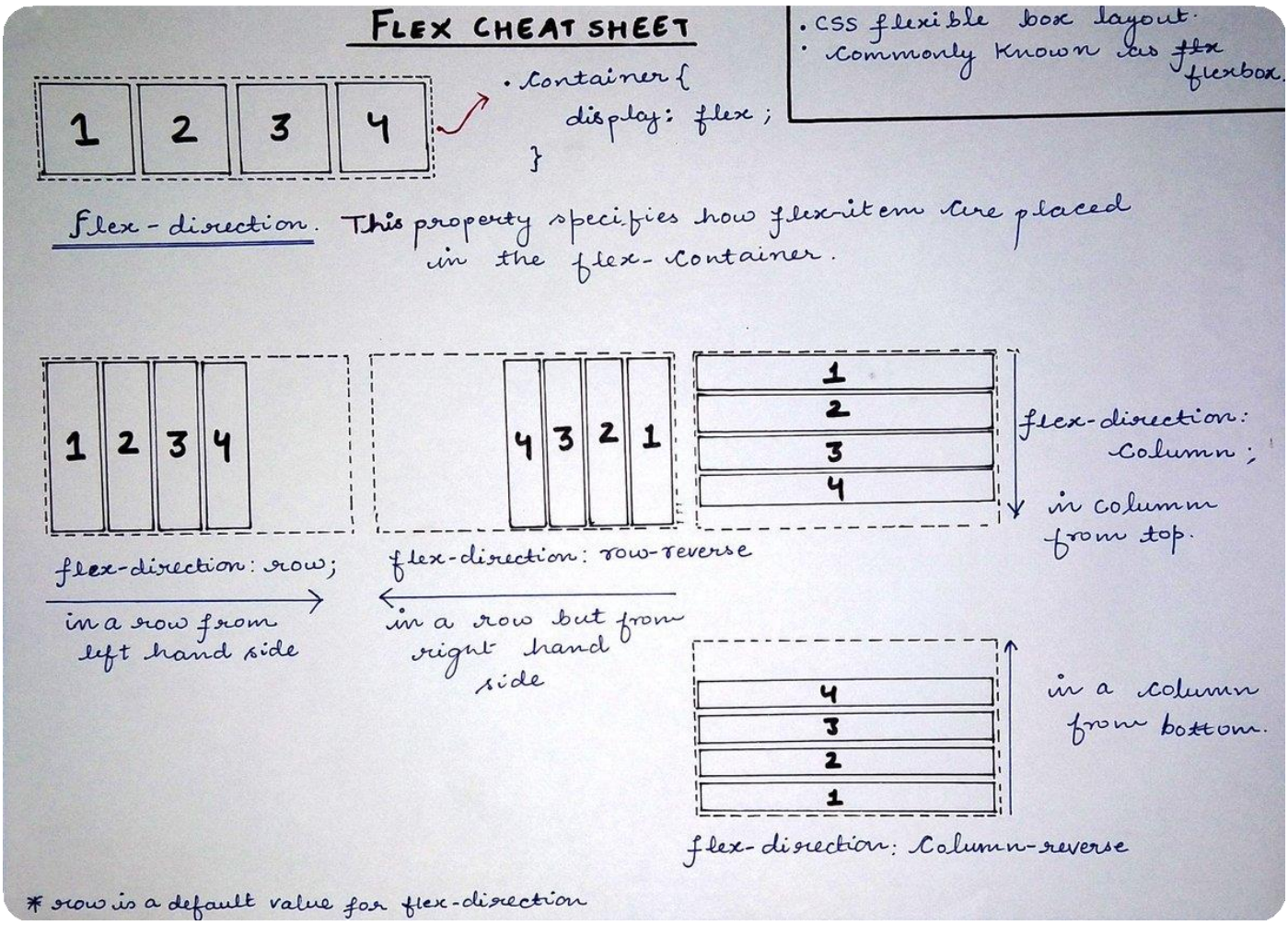



```
{align-items: end}
// align-items: center will align all grid items at center. at vertically
```

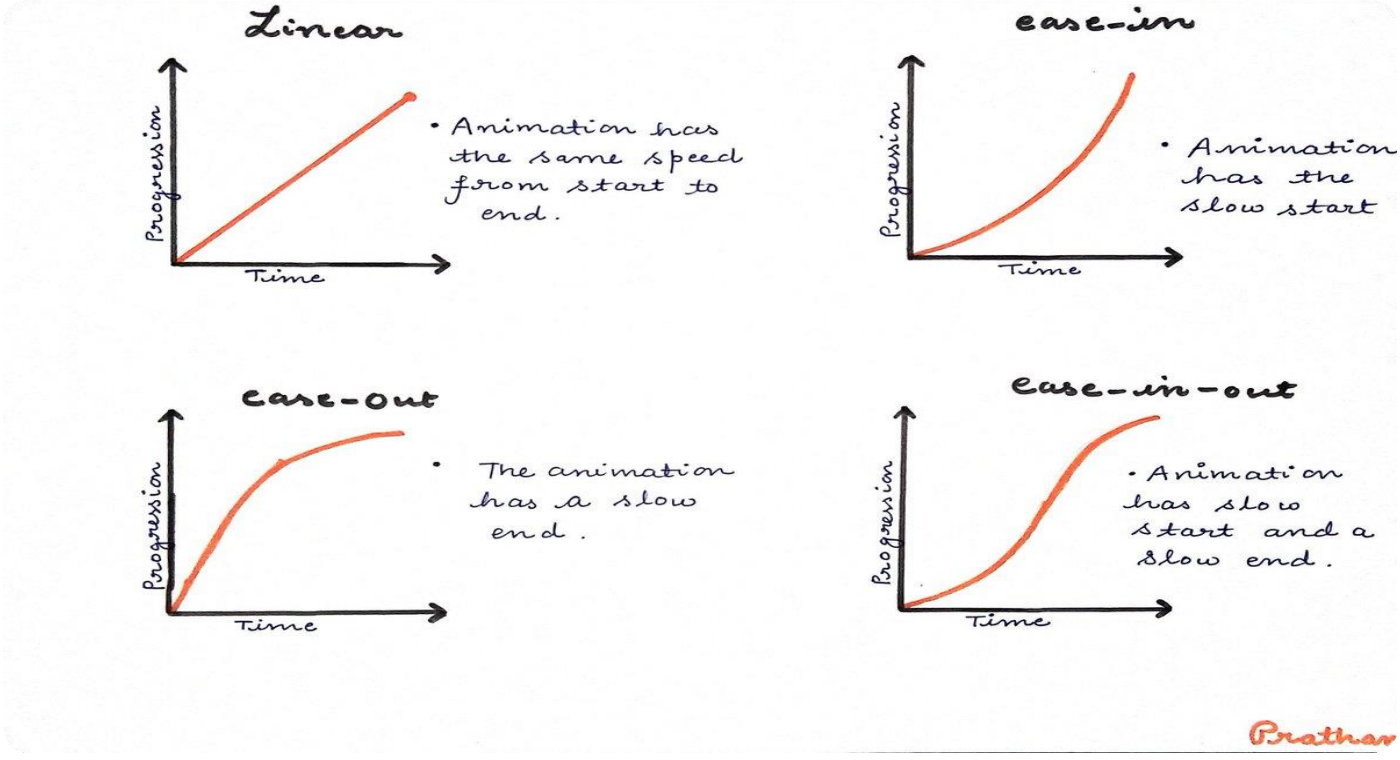


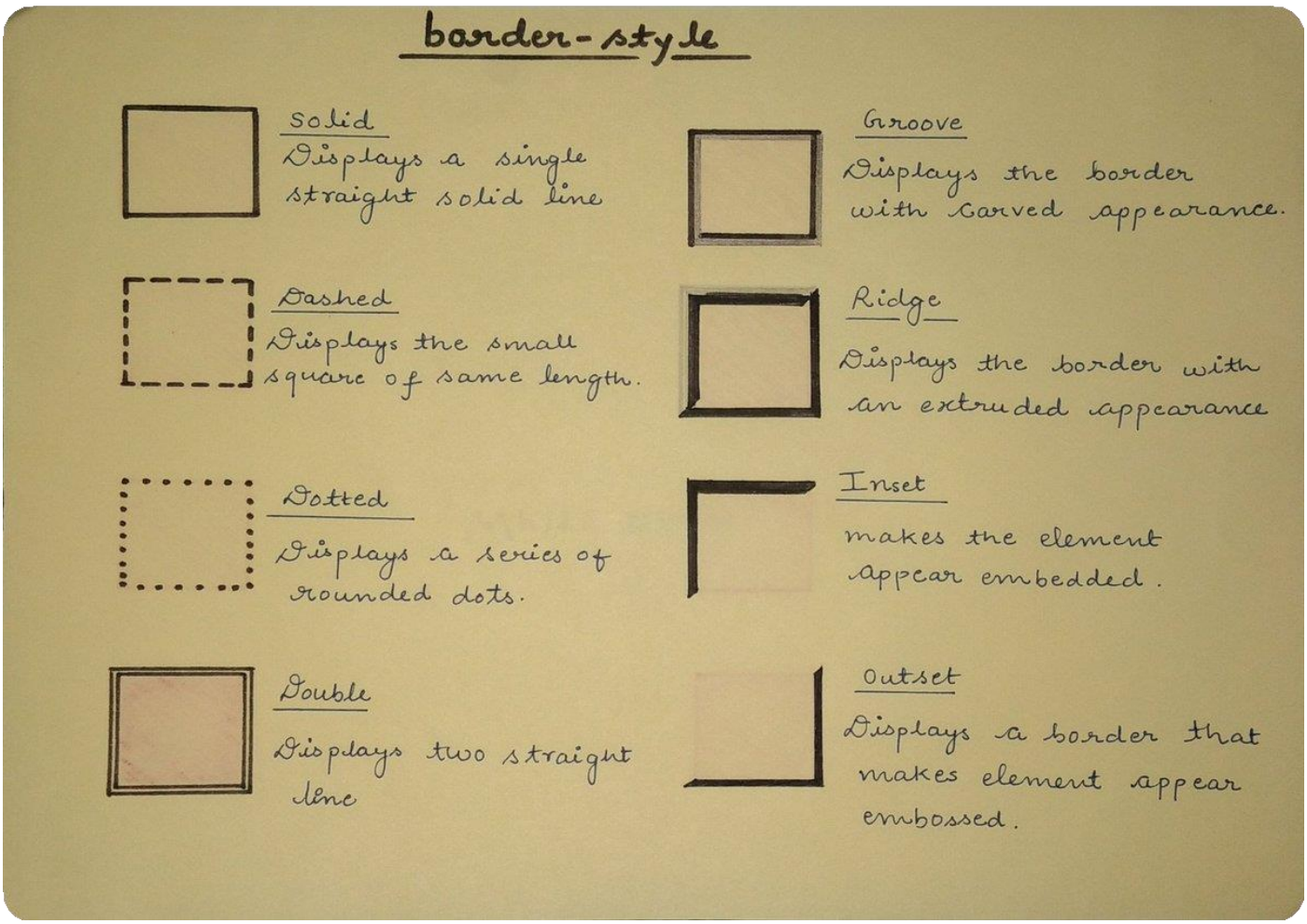
align-items: start

// It will align vertically.

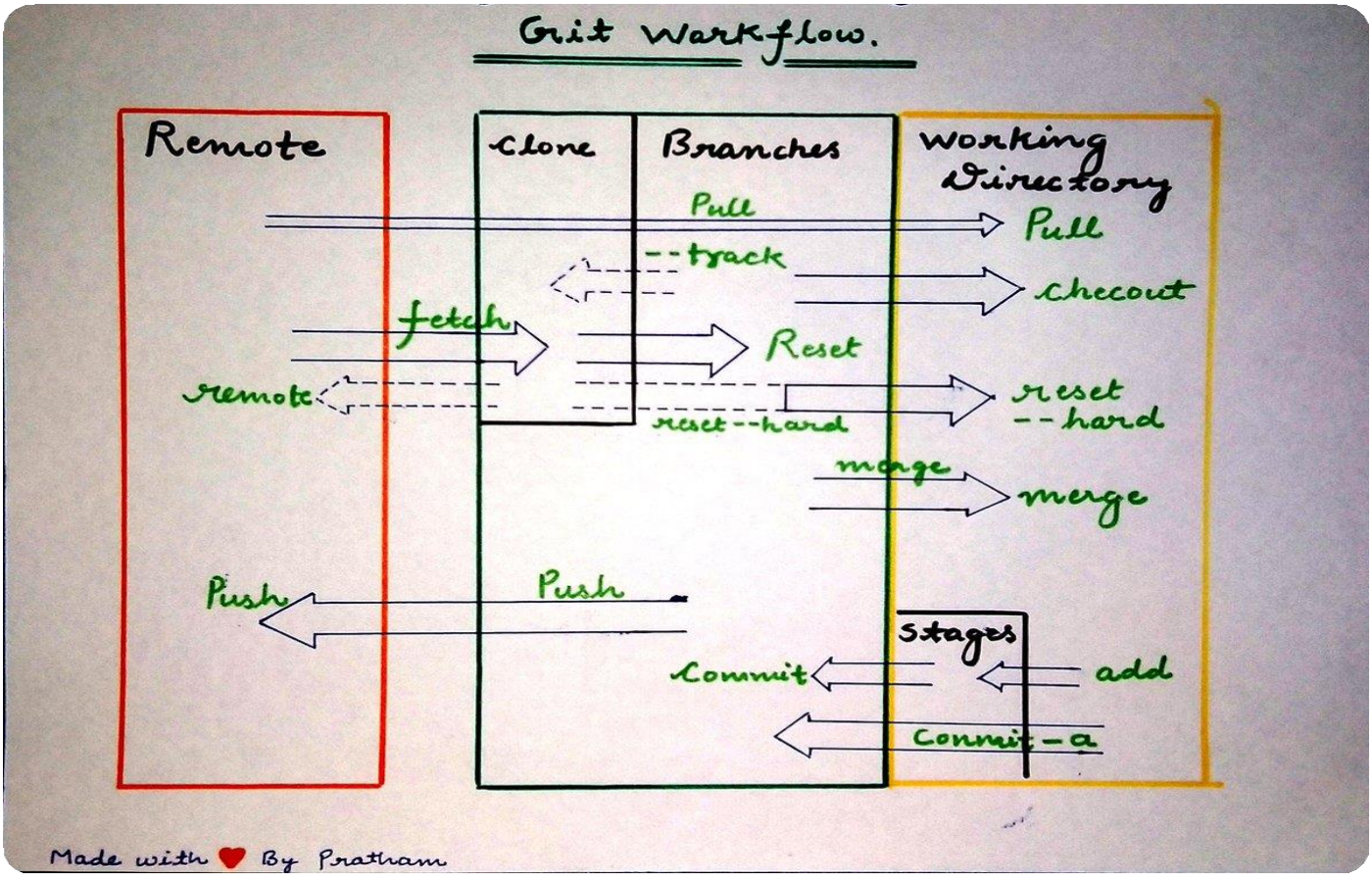


10. animation-timing-function

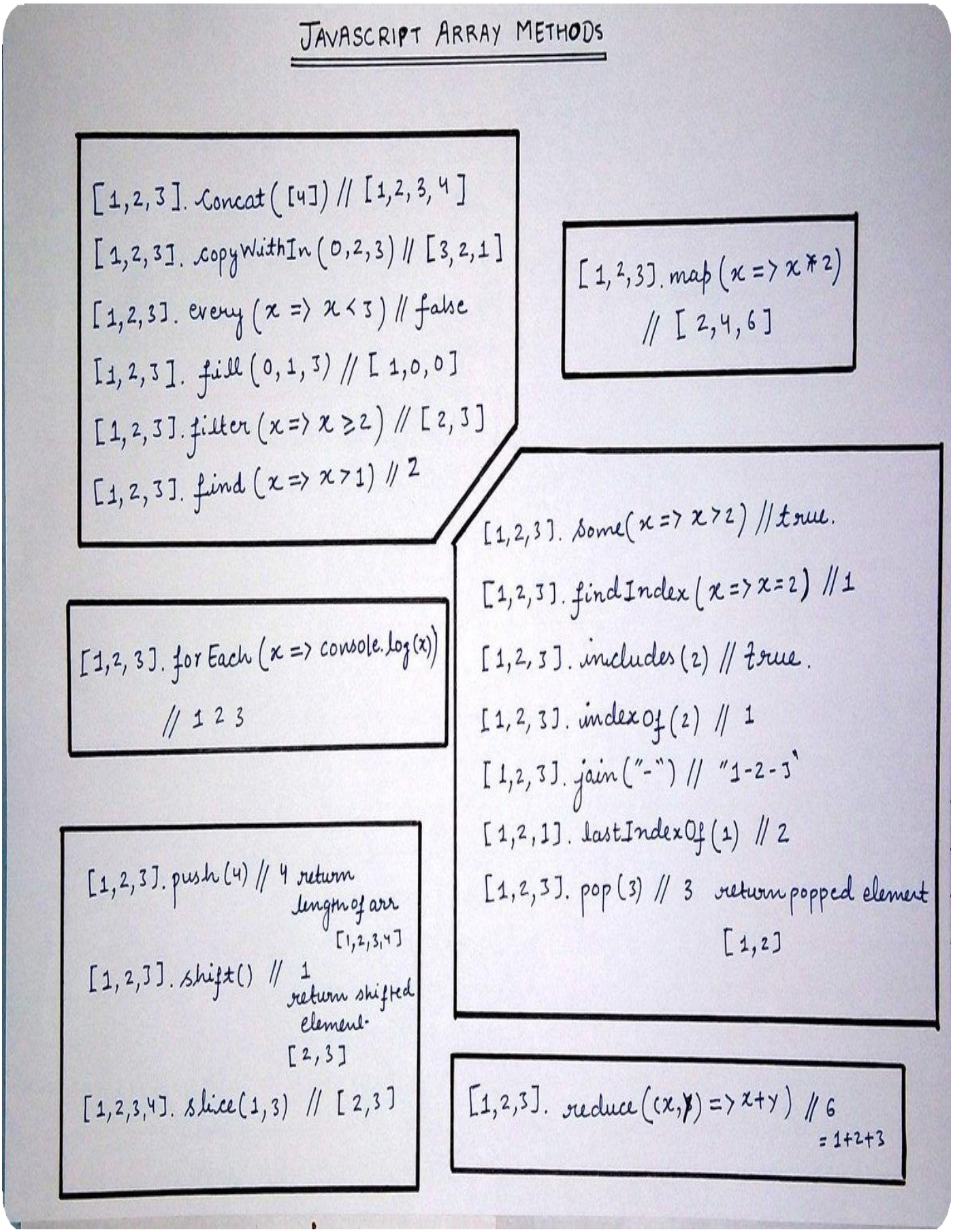




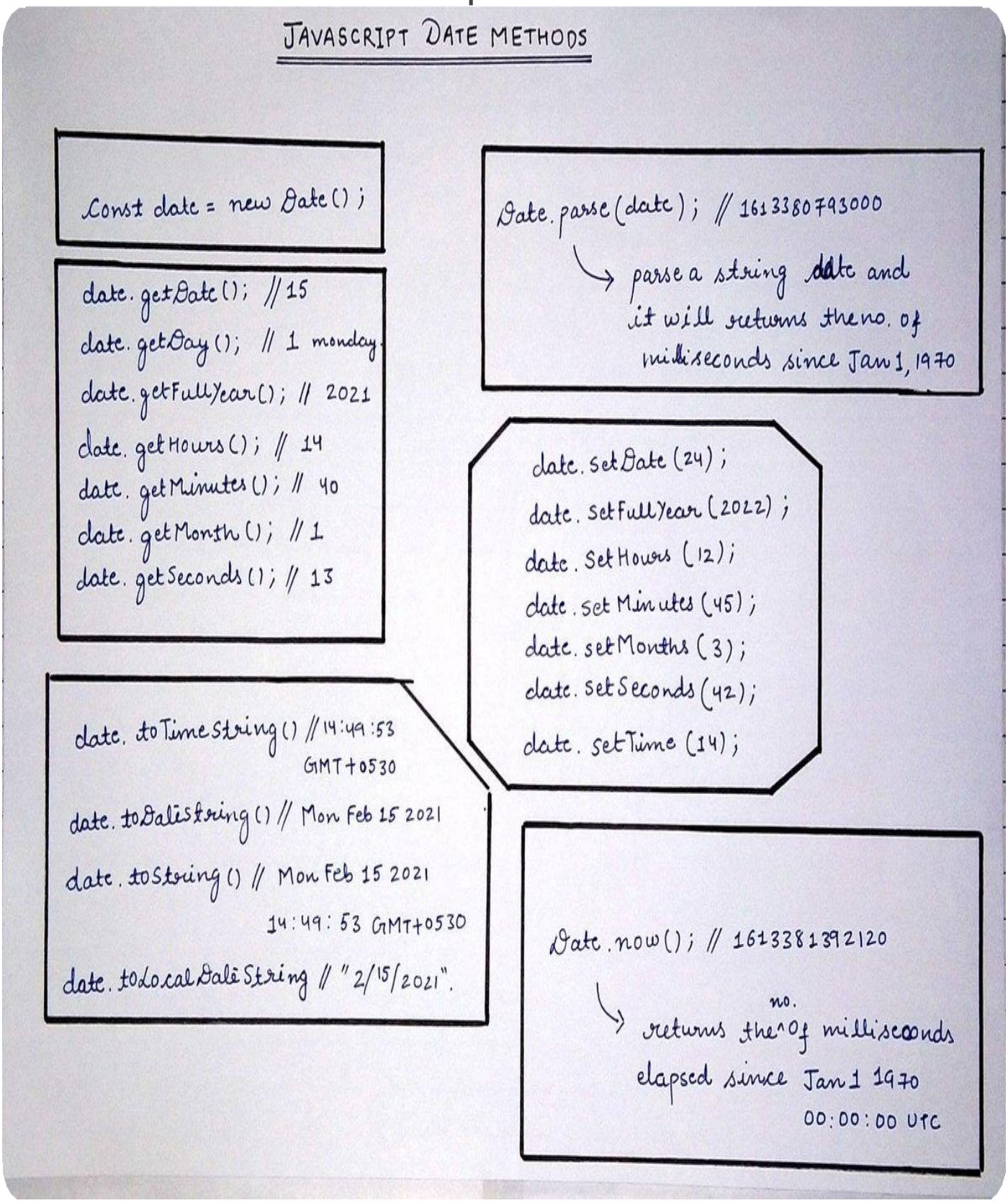
11. Git workflow



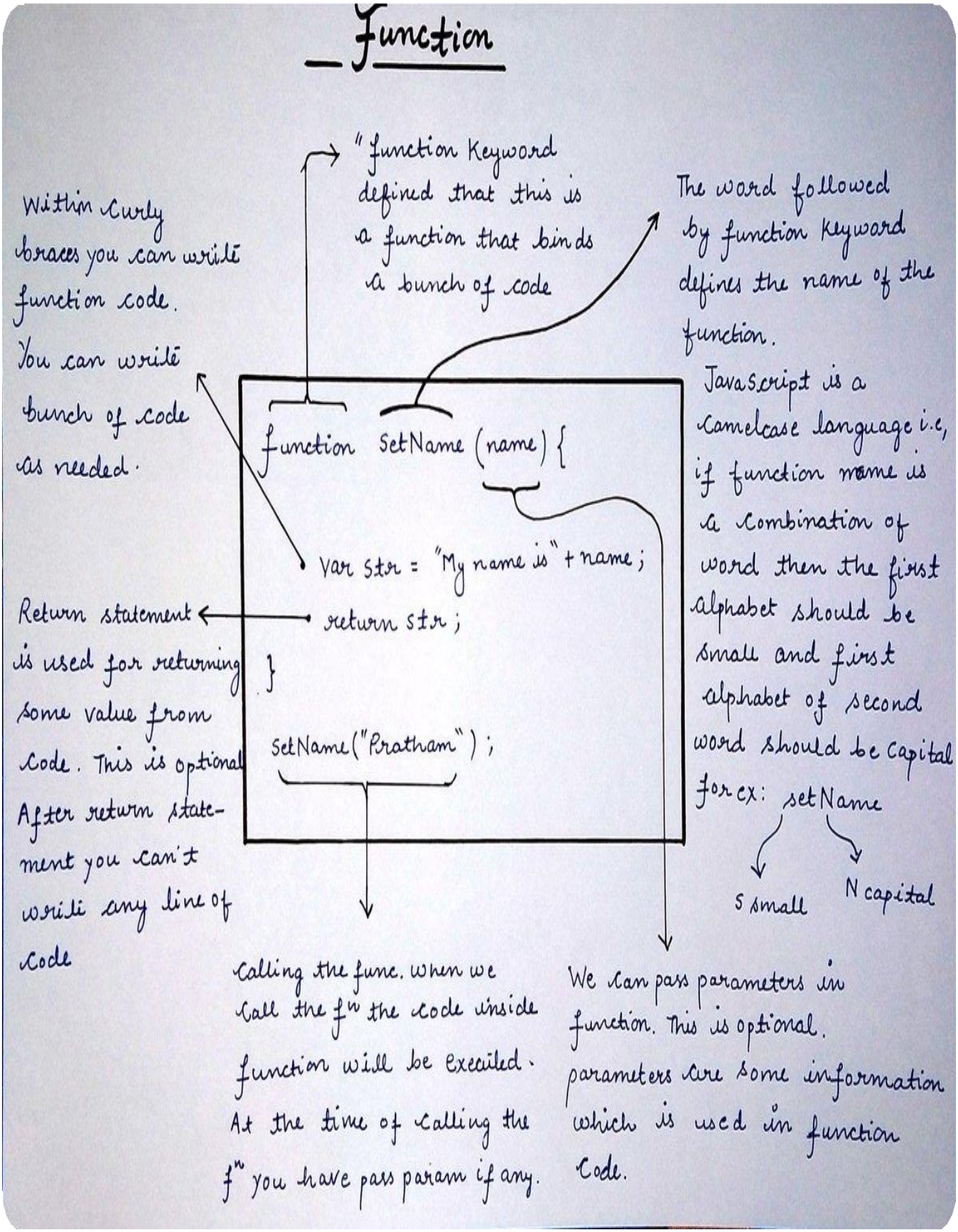
12. JavaScript Array methods cheat sheet



13. JavaScript Date methods



14. What is a functions



15. Arrow functions

Arrow function.

In arrow function you don't need to write the function keyword you don't even need to write the function name as well.

```
(a) => {
    return a+100;
}
```

parameter

- place an arrow sign between arguments and function body ({....})

```
(a) => a+100;
```

you can remove the braces and return keyword. It will work perfectly fine.

If in case there is a single parameter, then you can remove the argument parantheses.

```
a => a+100;
```

* If function body is there then return statement and parantheses ({...}) are required.

DOM

Getting elements.

```
document.getElementById("id");
// return the element with
   id = "id".
```

```
document.getElementsByClassName("class");
// returns the collections of element
   with class = "class"
```

```
document.getElementsByTagName("h1");
// return the collection of h1 element
```

```
document.querySelector(".class");
// return first element based on
   selector.
```

```
document.querySelectorAll(".class");
// return collection of element
   with class = "class";
```

Returns all elements with specified class name or tag name as a NodeList (collection of nodes).
In order to access specific node, you can use indexing. for ex,

```
document.getElementsByClassName("").[0]
```

DOM addEventListener() method

* addEventListener method attaches an event to the specified element.

Calling event listener method on button

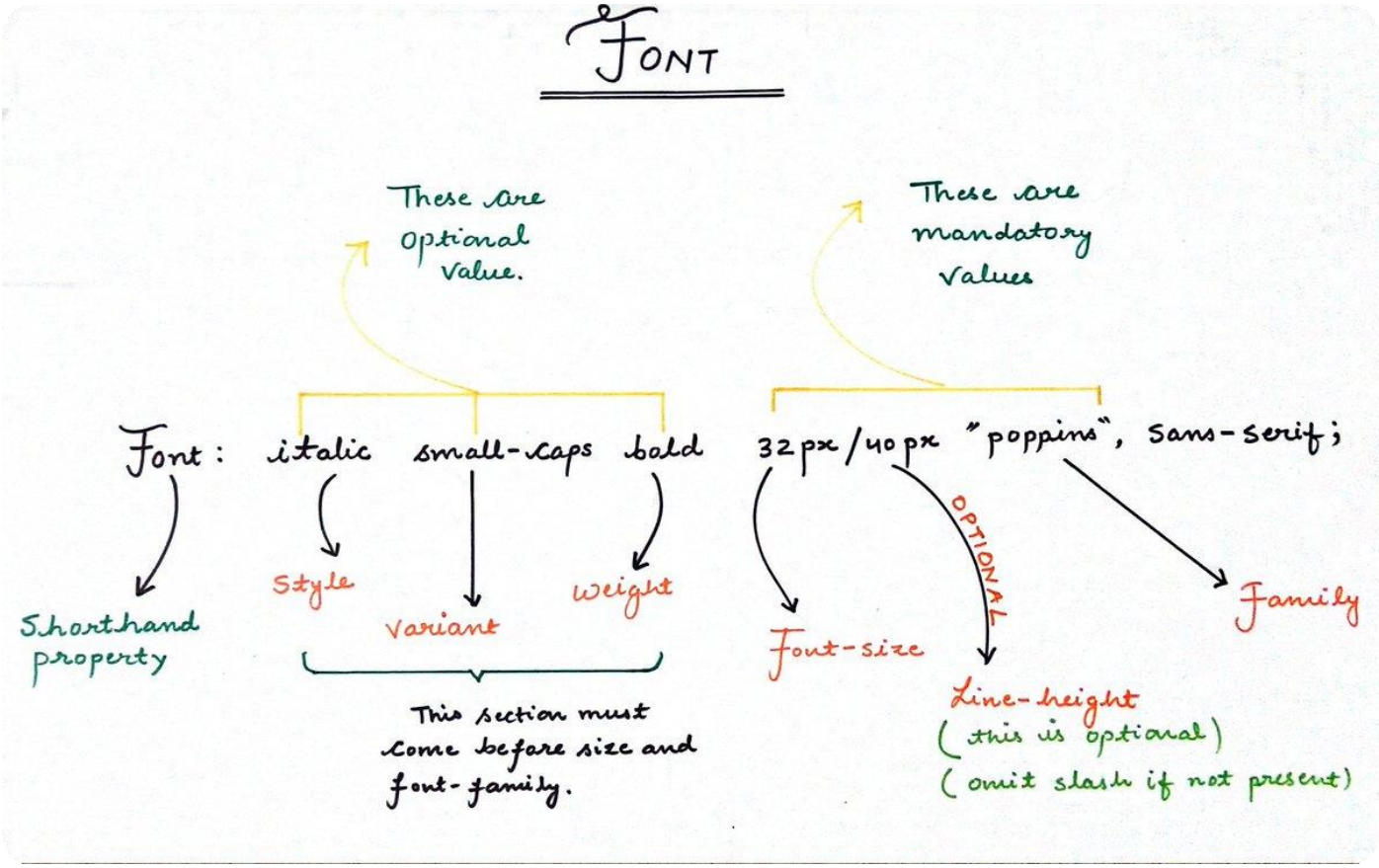
Getting button element from DOM

```
const btn = document.getElementById("btn")[0];
btn.addEventListener("click", function() {
    alert("clicked");
});
```

Passing "click" event as an argument. This event occurs when user clicks on the element.

This function will run when the event occurs. In this example, when button clicked, an alert will be displayed.

16. Font shorthand property



17. Array methods

```
[1,2,3].map(x => x * 2) // [2,4,6]
[1,2,3].filter(x => x<2) // [1]
[1,2,3].indexOf(2) // 1
[1,2,3].reduce((x,y) => x*y) // 6
[1,2,3].reverse() // [3,2,1]
```

```
['P','R'].concat(['A']) // ['P','R','A']
['P','R','a'].slice(1) // ['R','a']
['P','A','T'].splice(1,0,'R') // ['P','R','A','T']
['P','R'].join('-') // 'P-R'
['P','R','A','T','H','A','M'].lastIndexOf('A') // 5
```

Array Methods

```
[1,2,3,4].fill(0,2,4) // [1,2,0,0]
[1,2,3,4].find(x => x>3) // 4
[1,2,3].findIndex(x => x>1) // 1
[1,2,3].includes(2) // true
[1,2,3].some(x => x%2===0) // true
```

```
[1,2,3].forEach(x => console.log(x))
// 1,2,3
[1,2].push(3) // [1,2,3]
[1,2,3].pop() // [1,2]
[1,2,3].shift() // [2,3]
[1,2,3].unshift(0) // [0,1,2,3]
[1,2,3].every(x => x<5) // true
```

18. justify-items | align-items | justify-content | align-content

